

When collecting and analyzing anonymously submitted data, one must be aware of multiple considerations.

First, most anonymously submitted data has some form of bias, especially selection bias. In our case, students who have higher GPA's can be more likely to report them versus students with lower GPAs. Hence calculating metrics such as the mean GPA can be misleading as you are missing values from the full population and possibly over representing others. This is often an issue when trying to use anonymous survey data in analysis. Most respondents are either very happy, or very unhappy; those who are indifferent tend to not respond at all. In these cases, you generally see a skewed or bimodal distribution, when in reality if we had the full population represented it would be somewhat Gaussian.

A second consideration is that of correctness. Since the website allows users to type in the GPA value, some typos will make it into the data set. This is where outlier analysis can be helpful to remove these bad data points when they arise.

Because of these issues, I would treat the results as descriptive of the submission platform rather than as a true benchmark for all applicants. Some of the responses were a little surprising at first, but after I thought more carefully about the sampling process. The results differ from official standards because standards are usually based on broader, more systematic, and more validated populations, while this dataset reflects a narrow slice of people who decided to report their own outcomes and scores.

The GRE example is a good example. If the average quantitative score in a broader 2023 population was around 157, but the grad school entries in this dataset were closer to 165, that probably does not mean the overall applicant pool suddenly became dramatically stronger. More likely, the site attracts applicants targeting highly quantitative programs, applicants who feel more confident sharing strong scores, or applicants from especially competitive schools and fields such as computer science and engineering. The average could also be pushed upward because only some users report GRE scores at all, and those who omit them may be systematically different from those who include them. In other words, selection bias, reporting bias, field mix, prestige concentration, and possible data quality issues can all shift the mean well above the official standard. That is exactly why anonymous self-reported data can still be useful for exploratory analysis, but it should be interpreted cautiously and not treated as definitive results.